

Questions and ideas from deceased colleagues that help us carry on

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Abstract

Having been involved, over the past 30 years, in several conferences on the use of computer algebra systems in mathematics education (notably, the TIME, USACAS, T³IC conferences and the Education session of the ACA conferences), I have met several colleagues from whom I have learned a lot. The recent unexpected deaths of some of them, within only 15 months of each other, have upset many of us. An original way to honor their memory is to show how, starting from their personal mathematical concerns – quite different – this can lead to an attempt to answer the following question: how, surrounded today by all this technology and artificial intelligence, can we continue to teach mathematics to the current clientele of students? The examples presented will be for some mathematics courses that engineering students must take.

1 Introduction

Imagine, in the middle of spring 2023, José Luis Gálan García informs us of the death of Eugenio Roanes-Lozano, barely 60 years old and in perfect health. A few months later, David Jeffrey writes to me that he and David Stoutemyer are doing everything they can to recover Rubi's files since Albert Rich, diagnosed with Creutzfeldt -Jakob disease, only has a few days to go. And in the summer of 2024, Josef Böhm – who had participated in ACA 2023 in Warsaw – dies of cancer that he had courageously fought for several years.

A beautiful tribute to Eugenio-Roanes Lozano, professor at Complutense University of Madrid who passed away on April 16, 2023, was from Michael Wester : in fact, the Computer Algebra in Education session of ACA 2023 was dedicated to Eugenio. Among his many publications in symbolic computation, education, and transportation engineering, I must mention his very last presentation during the online session of ACA2021 and the paper [1]. If Eugenio asked the question of whether one can bring one's calculator to class, thinking mainly of an elementary or secondary school class, the same question arises at the university level — especially in an engineering school. ÉTS requires each of its undergraduate students to purchase a symbolic calculator (TI-Nspire CX

II CAS), which also comes with the computer software version. It is therefore in our own context that we will attempt to answer the question posed by Eugenio.

I will always remember how lucky I was when Albert Rich, who came to an ACA conference on a rare occasion (ACA2001 in Albuquerque), showed me how the next version of *Derive* was to contain a "Show Step Simplifications" for the computation of indefinite integrals. With the end of *Derive* in 2007, Albert continued this work for 15 years in the framework of his Rule-based Integrator system Rubi. Almost 40 years of presence on the market of computer algebra systems have not succeeded in changing the integral tables in calculus textbooks. By showing some rules of Rubi (via Mathematica), we will try to show that Eugenio's question makes sense if we adapt our way of presenting integration techniques to the Rubi way. Sadly, Albert's passing on August 11, 2023, came suddenly and put an end — at least for now — to a future version of Rubi.

The Education session of the ACA 2025 conference is dedicated to Josef Böhm. The website session contains a link that describes Josef's involvement well. Josef taught in Austria for many years in secondary schools and the use of technology was at the very heart of the concerns of the *Austrian Center for the Didactics of Computer Algebra* (ACDCA). Publishing the *Derive Newsletter* (DNL) from 1991 until his death showed that Josef was a proud user of technology and that he saw no contradiction between its use and paper and pencil. Josef was very interested in using graphs, sliders and iterations to illustrate many concepts, but it was questions about symbolic computation that led to several email exchanges over the past few years. Especially about symbolic integration!

2 Thank you Eugenio: I will use a two-part exam

Consider the following situation: We want engineering students to be able to use a computer algebra system not only for their homeworks but also for answering certain exam questions where such software is required. Of course, we want certain basic manipulations to always be performed by hand, without a machine. And we don't want students to have access to the internet during the exam. What should we do?

At ÉTS, we decided several years ago that using computer algebra systems should be an integral part of the exams our students must take. Our competitors will say that this is due to our special clientele, whose admission they would have refused... If there is any truth in this, we must admit that our students have the opportunity to tackle application problems that are practically impossible to solve without the assistance of a symbolic calculator. Aware that we must always achieve basic objectives regarding assessment, the present situation in the case of an exam in a mathematics course at ÉTS is no-calculator allowed for the first part of the exam (only a personal summary of 2-3 sheets) and the calculator (CAS handheld) in the second (with also the personal summary).

Here is the problem I have with the former situation. If we are in favor of banning technology during an exam, why then continue to allow the student to have in his possession a summary containing a large number of formulas, most of which are never used? Why not, instead, insist on a very small number of formulas and, while we are at it, force students to memorize a few of them? We are sorry to see students use tables during an exam — for example, table of integrals in a calculus course, table of Laplace transforms in a differential equations course — containing many formulas when a small number would be entirely appropriate. We are missing an opportunity to pay attention to *pattern recognition*, to the *symmetry* that may exist between certain formulas. For this

reason, here is my position — far from being shared by some of my colleagues. This is what I do for each of my midterm exams (At ÉTS, midterm exams are conducted by the professor who teaches the group, and the documentation permitted for the first part without a calculator may therefore vary from one group to another. However, since the final exam is the same for all groups of the same acronym, a personal summary is always permitted during the first part without a calculator).

- A first part of the exam without a calculator or personal summary ("nothing allowed"). The questionnaire may include some reminders. My decision to prohibit all documentation during the first part of the examination is to force the retention of certain definitions. This part of the exam accounts for approximately 35% of the total midterm exam.
- And a second part of the exam with a symbolic handheld calculator and a personal summary of 2-3 sheets. We are speaking of a CAS handheld: at ÉTS, this is TI-Nspire CX II CAS. This part of the exam accounts for approximately 65% of the total midterm exam.

So my personal answer to Eugenio's question is: yes, you can bring your *CAS handheld calculator* for the second part of the exam (you will also have to know how to use it !). And bring nothing but a pencil, an eraser, and a ruler for the first part of the midterm exam. If any special formulas deemed difficult to remember are required, they will be inserted into the statement (Of course, I would like this to become the standard not only for the midterm exam but for the final exam !).

But regardless of whether we disagree on the documentation permitted for the first part of the exam, we did agree that it should be written "symbolic calculator" and not "simple graphic calculator." Before giving examples that are completely true to the reality of my own institution, it is worth reminding those who are not aware of the decisions that led ÉTS to choose a symbolic calculator from the company Texas Instruments in 1999.

- In the early 1990s, some colleagues and I found it very useful to illustrate certain concepts in our math courses to future engineers using software such as Derive and/or Maple. We would bring a computer to class for a few minutes (with the required overhead projector!). The students remained passive, unable to do the same thing. At best, we would occasionally bring them to the computer lab.
- The situation changed greatly when the legendary TI-92 came onto the market in 1995. At ÉTS, some students who had calculators such as the TI-83 or HP-48 were becoming disadvantaged compared to those who had purchased a TI-92. Being involved with the Derive community since 1994, I was happy to see that a "small" Derive was finally available inside a CAS handheld !
- With the arrival of the TI-92Plus (followed by Voyage 200) and the desire to put everyone on the same footing while wanting to use the symbolic capabilities of this kind of machine, we decided to adopt TI handheld technology on all the ÉTS campus for the undergraduates students : the symbolic handheld calculator became mandatory starting in September 1999.
- And, in 2011, the transition to TI-Nspire CAS providing the dual-platform calculator/software greatly improved things. The calculator is allowed in part of the exam but never the software version on a computer. Moreover, the computer is prohibited during mathematics exams: no internet.

So let's give some examples of the kind of questions we can ask for each part of the exam.

2.1 Examples for an exam in linear algebra

Many years ago, we were computing determinant of a 5×5 square matrix by hand and spending times of reducing matrices using elementary row operations. In the end, several students had difficulty identifying the pivot columns. We still believe that for a *small* linear system, some basic manipulations should be made by the students and, in other cases, a CAS handheld can be used by the students but they will have to understand the problem.

2.1.1 Example 1 : nothing allowed

Find valid parametric equations for the line of intersection of the following planes using two different approaches : the Gauss-Jordan elimination method and the vector method (cross-product stuff):

$$\begin{cases} 2x + 4y - 1z = 1 \\ x + 3y - 2z = -2 \end{cases}$$

In other words, for the first method, a student needs to perform the row operations on the augmented matrix, to find the free variable and write the answer in parametric form. For the second method, the student needs to be able to compute by hand – perhaps the definition of the vector product could be recalled in the questionnaire – the cross product of the vectors $[2, 4, -1]$ and $[1, 3, -2]$ yielding a parallel vector to the line of intersection of the two planes and the student also needs to find a point lying on both planes, setting for example $z = 0$ in both equations, and solving the linear system of two equations in two unknowns.

2.1.2 Example 2 : CAS handheld and personal summary of 3 sheets

For what values of the parameter α does the following system of linear equations have a unique solution, no solution, or an infinity of solutions? The system is $\mathbf{B}\mathbf{X} = \mathbf{K}$ where

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & -3 \\ 2 & \alpha & -1 \\ 1 & 2 & \alpha \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{K} = \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}.$$

Here the student should avoid using a command such as `rref()` because one (or more) value could be missing. What remains important is to recall that a square linear system of equations has a unique solution if and only if the determinant of the coefficients matrix is not equal to 0. A CAS command will easily find that the determinant of matrix $\mathbf{B}$ is 0 when $\alpha = -5$ or $\alpha = 2$ but row reduction will "forget" one value. This is the case with common computer algebra systems : Figure 1 for Derive, Figure 2 for Maple and Figure 3 for Nspire. One interesting point is the fact that, having access to a CAS handheld during the second part of the exam, the students can define the matrix $\mathbf{B}$ as a function of α and use a command such as `rref()` with the values inserted before the operation. Figure 3 shows that for $\alpha = -5$, there is no solution; for $\alpha = 2$, there is an infinity of solutions and for all other values, there is a unique solution.

$$\#1: \quad \text{ROW_REDUCE} \begin{bmatrix} 1 & 0 & -3 & -3 \\ 2 & \alpha & -1 & -2 \\ 1 & 2 & \alpha & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -\frac{3 \cdot (\alpha + 1)}{\alpha + 5} \\ 0 & 1 & 0 & \frac{4}{\alpha + 5} \\ 0 & 0 & 1 & \frac{4}{\alpha + 5} \end{bmatrix}$$

Figure 1: Using `row_reduce()` on a symbolic matrix with Derive

$$\text{LinearAlgebra}[\text{ReducedRowEchelonForm}] \left(\begin{bmatrix} 1 & 0 & -3 & -3 \\ 2 & \text{alpha} & -1 & -2 \\ 1 & 2 & \text{alpha} & 1 \end{bmatrix} \right);$$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3(1+\alpha)}{\alpha+5} \\ 0 & 1 & 0 & \frac{4}{\alpha+5} \\ 0 & 0 & 1 & \frac{4}{\alpha+5} \end{bmatrix}$$

Figure 2: Using `rref()` on a symbolic matrix with Maple

$$\begin{aligned} \text{mat}(\alpha) &:= \begin{bmatrix} 1 & 0 & -3 \\ 2 & \alpha & -1 \\ 1 & 2 & \alpha \end{bmatrix} \rightarrow \text{Done} & \quad & \text{rref} \left(\text{augment} \left(\text{mat}(\alpha), \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix} \right) \right) \rightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{12}{\alpha+5} - 3 \\ 0 & 1 & 0 & \frac{4}{\alpha+5} \\ 0 & 0 & 1 & \frac{4}{\alpha+5} \end{bmatrix} \\ \text{zeros}(\det(\text{mat}(\alpha)), \alpha) &\rightarrow \{-5, 2\} \\ \text{rref} \left(\text{augment} \left(\text{mat}(-5), \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix} \right) \right) &\rightarrow \begin{bmatrix} 1 & 0 & -3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \quad & \text{rref} \left(\text{augment} \left(\text{mat}(2), \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix} \right) \right) \rightarrow \begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & 1 & \frac{5}{2} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Figure 3: Using `rref()` on a symbolic matrix with Nspire

2.2 Examples for an exam in ODEs

Many years ago, students were solving a lot of different types of ODEs by hand. They were learning tricks to solve different kinds of first orders ODEs : linear, separable, Bernoulli, homogeneous, exact, non-exact with integrating factor (but almost all application problems only involved linear or separable equations). For the linear second order, constant coefficients ODEs, the important method of underdetermined coefficients is still quite important but the one of variation of parameters requires the computation of integrals and, as long as mathematics engineering is concerned, a mechanical or electrical system will contain right hand input functions for which the undetermined coefficients method works well.

2.2.1 Example 3 : nothing allowed

I emphasize especially the recognition of form and the first order differential equations to be solved are of the linear or separable type whose computation of integrals does not require any table. Here is an example. Consider the first order ODE with initial condition :

$$\frac{dy}{dx} = y + 3, \quad y(0) = 2.$$

Before trying to solve it, give sufficient conditions ¹ that assure us that there does indeed exist a unique solution defined in the neighborhood of the point $(2, 0)$.

Now, solve it by considering it as *linear* differential equation and give your answer as an explicit function of y . Finally, solve it again as a *separable* differential equation and give again your answer as an explicit function of y . Let us recall that the formula for solving the *linear* differential equation $\frac{dy}{dx} + p(x)y = q(x)$ is given by

$$y = \frac{1}{\mu(x)} \left(\int \mu(x) q(x) dx + C \right) \quad \text{with} \quad \mu(x) = e^{\int p(x) dx}.$$

2.2.2 Example 4 : CAS handheld and personal summary of 3 sheets

Typical questions involve finding the steady-state current in an RLC circuit whose source is a sinusoidal function, finding its period and amplitude, and reproducing the graph, over a given interval, in the exam booklet. Another typical question — and this shows that the second part of an exam can very well contain a significant amount of paper and pencil work — is to use power series method to estimate with *accuracy* the value of the solution $y(x)$ at $x = 1.9$ for the ODE

$$(x^2 + 4) \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + y = 0, \quad y(0) = 2, \quad \frac{dy}{dx}(0) = -3.$$

They will need to use a series solution $y(x) = \sum_{n=0}^{\infty} a_n x^n$, $a_0 = 2$, $a_1 = -3$ and find the recurrence formula (the CAS handheld does not do this for them). Then they will need to use a sufficient order partial sum to compute **with accuracy** the value of $y(1.9)$ because the interval of convergence of the series will be $-2 < x < 2$. And if they want to validate their answer, they can transform the

¹This question tests if the students understand the theorem of existence and unicity of solutions

problem into a first order system and use the built-in RK method of Nspire : setting $\frac{dy}{dx} = z$ will transform the second order problem into the first order following system :

$$\begin{cases} \frac{dy}{dx} = z, & y(0) = 2 \\ \frac{dz}{dx} = \frac{3z - y}{x^2 + 4}, & z(0) = -3 \end{cases}$$

Take a good look at the previous system of differential equations. Students will be able to estimate the value of the solution $y(x)$ at the point $x = 1.9$ by plotting the graph generated by the RK method. Since there are three variables (x , y , and z), they must therefore select the axes in a 2D graph window. This is something we never did before using a symbolic calculator in class. Using a CAS handheld in the classroom is an opportunity to plot many different kinds of 2D graphics :

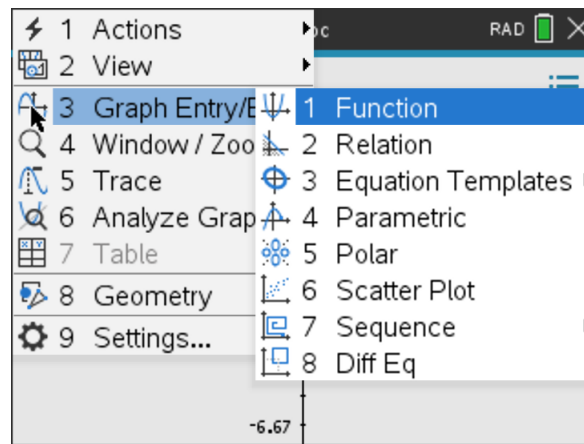


Figure 4: TI-Nspire CX II CAS handheld : list of 2D plot windows

3 Albert : thank you so much for showing us the *rules*

One of the main reasons to continue teaching different integration techniques is that it forces students to manipulate expressions, do algebra, use some important formulas (especially in trigonometry) and many other things. Far be it from us to no longer want our students to do this!

But in the case of users of mathematics such as future engineers, in the particular case of an institution where applications are a primary and conscious objective, continuing to teach integration techniques while ignoring the last 35 years of symbolic computation systems seems somewhat counterproductive to us. Worse still, our table books continue to use primitives with almost no justification of the origin of the result and remind the reader to add a constant of integration while the expressions to be integrated are possibly discontinuous on certain intervals. There is also the following discomfort: we flood students with integration formulas but we do not spend enough time on justifying the most common ones.

Adding, into a table of integrals, the domains where the integrand is continuous would produce huge tables, which would be unnecessarily complicated : but nothing prevents us, as a professor,

from inserting such a table, which we did during our presentation at ACA2024: see for instance table 1.

Take for example the expression $\frac{1}{a^2-x^2}$ where $a \in \mathbb{R}^*$. What antiderivative of this expression are we going to find if we look to some textbooks or use a particular computer algebra system or Wofram Alpha ? It is unlikely that it will be accompanied by the domain of definition of the primitive. We also said that it was time to use the hyperbolic arctangent function in our integral tables. While it is true that $\operatorname{arctanh}(x)$ is only real for $-1 < x < 1$, it is much easier to remember as an antiderivative of $\frac{1}{1-x^2}$ than those containing logarithms and, in addition, it offers a nice symmetry with the antiderivative $\operatorname{arctan}(x)$ of the expression $\frac{1}{1+x^2}$.

Another important point is that students will see that the imaginary parts cancel out when they choose the antiderivative $\operatorname{arctanh}(x)$ to calculate the area under the curve of the expression $\frac{1}{1-x^2}$ between $x = 5$ and $x = 6$.

Domain	Real antiderivative of $\frac{1}{a^2-x^2}$
$-a < x < a$	$\operatorname{arctanh}\left(\frac{x}{a}\right) = \frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right)$
$x > a$ or $x < -a$	$\frac{1}{2a} \ln\left(\frac{x+a}{x-a}\right) = -\frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$
$x > a$	$\frac{1}{2a} \ln(x+a) - \frac{1}{2a} \ln(x-a)$
$x \neq \pm a$	$\frac{1}{2a} \ln\left(\left \frac{x+a}{x-a}\right \right)$

Table 1: Four different antiderivatives of $\frac{1}{a^2-x^2}$ with their respective domain

So table 2 contains the formulas that, in my opinion, should be memorized by an engineering student. Using substitution and this simple table, we have enough to produce interesting questions for the exam where nothing is allowed. A more extensive table – influenced by Rubi – can be found in [2].

Let's not be afraid to repeat it. If we're teaching mathematics in an engineering school where every student has a symbolic calculator on their desk, we are convinced that :

- it's better to focus on a short table of integrals and – in our opinion – ask them to memorize it. Students will rapidly learn that it becomes easy to remind the formulas for the integral of $\frac{1}{x+a}$, e^{ax} , $\sin(ax)$ and $\cos(ax)$ where $a \neq 0$ is a constant.
- it's ridiculous to continue writing $\ln(|x|) + C$ as an antiderivative of the expression $\frac{1}{x}$ since we integrate over intervals and $\frac{1}{x}$ is not continuous on any interval containing the origin.
- we must quickly introduce them to hyperbolic functions so that they can then appreciate the symmetry between formulas (7) and (8) of table 2.

(1) $\int dx = x$	(2) $\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$
(3) $\int \frac{1}{x} dx = \ln(x)$	(4) $\int e^x dx = e^x$
(5) $\int \sin(x) dx = -\cos(x)$	(6) $\int \cos(x) dx = \sin(x)$
(7) $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) \quad (a \neq 0)$	(8) $\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \operatorname{arctanh}\left(\frac{x}{a}\right) \quad (a \neq 0)$

Table 2: Eight common integrals to remember

3.1 Examples in single variable calculus

As I wrote earlier, at ÉTS, even though we are using a CAS handheld, we agree that students should continue to be tested on computing integrals : techniques such as change of variable (substitution) and integration by parts are questions included in the first part of the calculus course. But I disagree when a table of integrals is given to the students. I would also include in the exam the integration of simple rational functions such as $\frac{ax+b}{cx^2+dx+e}$ since this is an excellent way to distinguish between completing the square (when the roots of the denominator are complex) and factoring the denominator in other cases. In other words, pattern recognition becomes important.

3.1.1 Example 5 : nothing allowed

Here is a list of integrals engineering students should be able to compute **with nothing allowed**. We suppose they think of using the linearity of the integral – something that is really important and seems to be forgotten by many students; they also need to have memorized the formulas of table 2, to know some elementary trigonometric identities and basic algebra. The professor can decide to recall the formula for integration by parts.

- | | |
|--|---|
| (a) $\int x^2(4 - 5x^3)^{12} dx$ | (h) $\int \frac{y + 20}{y^2 + 40y + 16} dy$ |
| (b) $\int e^{3w} \sin(e^{3w}) dw$ | (i) $\int \frac{x}{x^2 + 10x + 21} dx$ |
| (c) $\int \left(\pi x^2 - 7 + \frac{3 \cos(x)}{\sin(x) + 4} \right) dx$ | (j) $\int \frac{7x - 1}{x^2 + 8x + 25} dx$ |
| (d) $\int \frac{7x}{1 + x^4} dx$ | (k) $\int \frac{2 - x}{x^2 + 14x + 49} dx$ |
| (e) $\int (s^2 + 3)^2 ds$ | (l) $\int x e^{2x} dx$ |
| (f) $\int \frac{z^2}{1 - z^6} dz$ | (m) $\int \theta \cos(\theta) d\theta$ |
| (g) $\int_0^{\pi/4} 5 \cos^2(2x) \sin(2x) dx$ | |

Albert Rich once wrote : "Indefinite integration accounts for only a tiny fraction of all mathematical knowledge. However, based on my experience implementing muMath, Derive, and now Rubi, I see no reason why much of the rest of mathematics cannot be organized as a rule-based decision tree as well. Certainly much of analysis including equation solving, expression simplification, differentiation, summation, limits, etc. can be automated using this paradigm".

We don't need to use Rubi to apply this principle. Just take the previous questions (i), (j) and (k) of **Example 5** and replace the numerator in each with the number 1. By asking Derive and now Rubi to show their simplifications, we would see very clearly that everything begins with the analysis of the roots of the denominators: is this not an eloquent illustration of their pedagogical qualities? When you teach to students who a non negligible of them have forgotten the importance of the discriminant of a quadratic expression, showing them the "steps" is something you need to do.

3.1.2 Example 6 : CAS handheld and personal summary of 3 sheets

Many interesting antiderivatives provided by Rubi can be used to ask questions on integrals ! But before showing this, the handheld used at ETS (TI-Nspire CX II CAS) is quite powerfull but has not as many special functions already implemented into the "big" CAS (Maple, Mathematica for instance). This becomes an opportunity to ask students to compute by hands some integrals and also use their handheld to check the answer as soon as they will have defined the special function into their own calculator.

Here are some questions that would not only be original but would also take into account the acquired knowledge that symbolic systems and Rubi provide. With only a symbolic calculator like the current TI-Nspire CX II CAS (OS6.2 in May 2025), the student would have work to do but could use their "mathematical assistant"!

(a) Define the following function into your CAS handheld:

$$\text{Si}(x) = \int_0^x \frac{\sin(t)}{t} dx.$$

Then compute, in terms of Si, the exact value of

$$\int_3^5 \frac{\sin(2v)}{v} dv.$$

(b) For the following integral, check whether your symbolic calculator gives the same answer as the one displayed. If not, find the constant of integration by which the two antiderivatives possibly differ.

$$\int \sin^4(x) dx = \frac{\cos(4x) - 4 \cos(2x) + 3}{8}.$$

(c) For the following integral, check whether your symbolic calculator gives the same answer as the one displayed. If not, find the constant of integration by which the two antiderivatives possibly differ.

$$\int \frac{x^4}{(1+x)^6} = \frac{x^5}{5(x+1)^5} \quad (\text{for the interval of continuity, choose } x > -1).$$

(d) With Rubi, one of Albert's goal was to find the optimal antiderivative. A simple example of this could be question (f) of **Example 5**. Figure 5 and Figure 6 show how, from *Derive* to Rubi, Albert evolved his integration rules. All of these more than 6600 rules can be viewed on the Rubi website ([3]).

#1:
$$\int \frac{w^2}{1-w^6} dw$$

$\int -F(x) dx \rightarrow - \int F(x) dx$

#2:
$$- \int \frac{w^2}{w^6 - 1} dw$$

$$\int \frac{x^{n-1}}{b \cdot x^{2 \cdot n} - a} dx \rightarrow \frac{\text{LN}(\sqrt{b} \cdot x^n - \sqrt{a})}{2 \cdot n \cdot \sqrt{a} \cdot \sqrt{b}} - \frac{\text{LN}(\sqrt{b} \cdot x^n + \sqrt{a})}{2 \cdot n \cdot \sqrt{a} \cdot \sqrt{b}}$$

#3:
$$- \frac{\text{LN}(w^3 - 1)}{6} + \frac{\text{LN}(w^3 + 1)}{6}$$

Figure 5: *Derive* was not using arctanh !

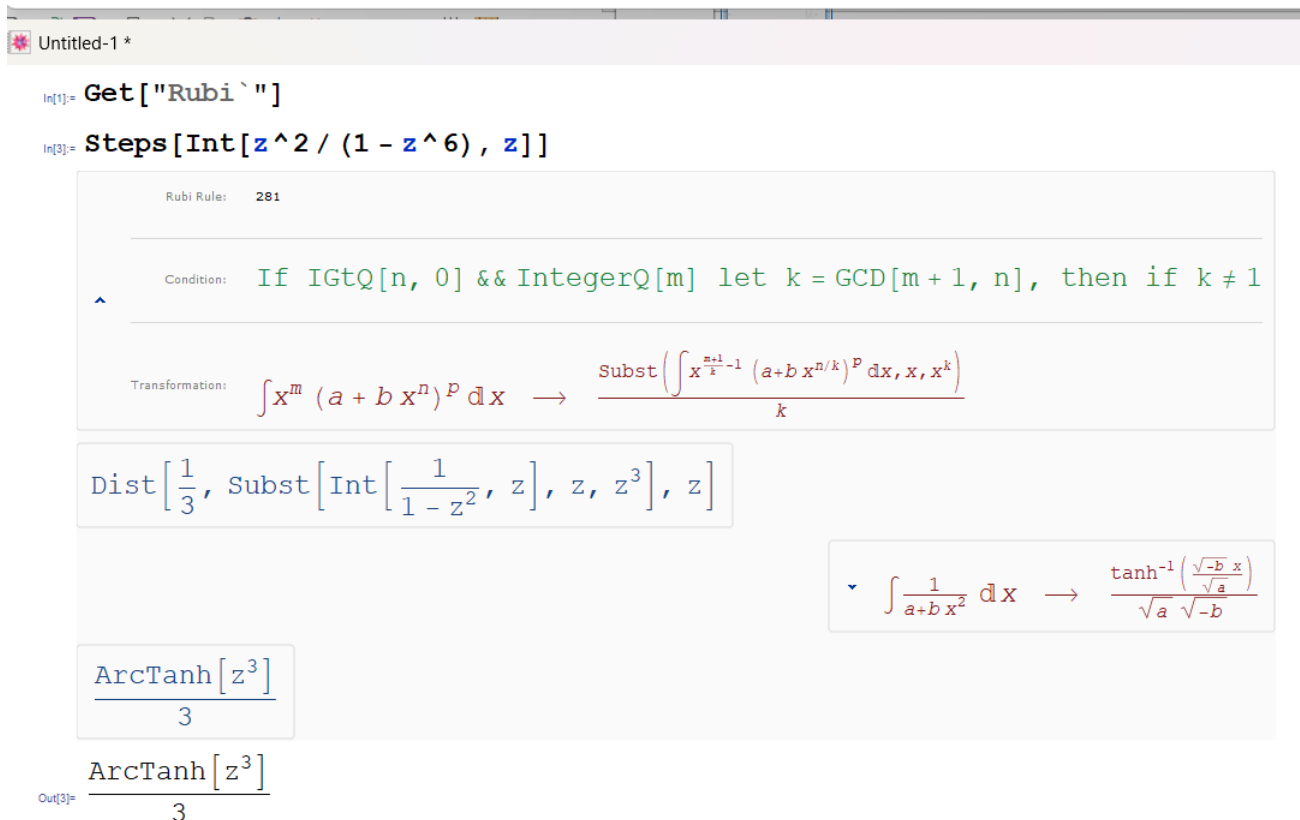


Figure 6: Rubi is using arctanh !

Albert was much more of a loner. For those of you who didn't know him — or barely knew him — check out this summary by David Stoutemeyer and David Jeffrey presented during the Integration and ODEs session at ACA2024. Without explicitly admitting it, Albert Rich was very shocked by Derive's abrupt end in 2007. And when Josef Böhm was writing to him to ask how Derive did this or that, Albert replied that he no longer used Derive and that all his work was now with Rubi! That's how Josef started questioning me about the symbolic computation aspect, especially since he didn't have Mathematica installed on his computer and therefore couldn't use Rubi.

4 Dear Josef, I tried my best to answer your questions

In Austria, since the early 1990s, along with Helmut Heugl and Bernhard Kutzler, Josef Böhm has been one of the pioneers of the use of Derive software in mathematics teaching. While recent years have shown a decline in the use of computer algebra systems – a predominant place being occupied by DGS, namely Geogebra – it must be admitted that Josef Böhm has not ceased to be one of the most loyal European users of Derive and TI-Nspire.

Among the twenty or so texts of mine that Josef published in the Derive Newsletter (between issues 69 and 132), several came from presentations made alone or in teams with colleagues from

ÉTS. I am still convinced that some of these texts could have been published if the writing had been more professional and others were simply responses to questions asked by Josef about simplifications made by software that he did not understand. See for instance [4]. A unique way for me to pay tribute to Josef is to make a short list of his latest questions regarding automatic simplifications made by software. From memory, this began at the TIME2012 conference, where a discussion on the theme *Derive is still special* was organized by José Luis Gálan García, the late Carl Leinbach, Josef, and myself. It then continued through email exchanges and in several issues of DNL.

Josef had long since retired and had left the teaching profession. For him, using math software was a normal thing when it came to solving problems without administrative or academic constraints. We will therefore conclude this text by providing examples where, despite the tools Josef had at his disposal, intermediate steps eluded him in understanding the displayed answer. This provided us with valuable exchanges.

4.1 The problem of simplifying nested radicals

Over the years, problems of different natures were actually connected by the fact that a simplification of nested radicals was required to obtain the answer displayed by the software. Here are some examples that gave Josef and me a lot of work!

4.1.1 Example 7 : everything allowed but try do show it

A nice chapter of the book edited by Michael Wester (see [5]) contains explanations how Derive was doing to try to simplify some nested radicals. These can easily appear with problems as simple as trying to show why

$$\sin(15^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}$$

or solving quadratic formula. Let's chose the latter found in [5]. Suppose you want to solve the following quadratic equation

$$x^2 + 6x - 4\sqrt{5} = 0.$$

Using the quadratic formula, we get

$$x = \frac{-6 \pm \sqrt{36 + 16\sqrt{5}}}{2} = -3 \pm \sqrt{9 + 4\sqrt{5}}.$$

As Figure 7 shows, every of our usual CAS yields the same answer which seems different from the one above. I told Josef that rules have been added (in Derive by Jeffrey/Rich) for simplifying nested radicals; one of these rules is the following that helps understanding the two solutions of the quadratic equation :

$$x > y > 0 \Rightarrow \sqrt{x \pm y} = \sqrt{\frac{x + \sqrt{x^2 - y^2}}{2}} \pm \sqrt{\frac{x - \sqrt{x^2 - y^2}}{2}}.$$

Being a CAS enthusiastic user, Josef convinced himself of this by using many "experimental" approaches : namely by defining the function of two variables

$$f(x, y) = \sqrt{x + y} - \sqrt{\frac{x + \sqrt{x^2 - y^2}}{2}} - \sqrt{\frac{x - \sqrt{x^2 - y^2}}{2}}$$

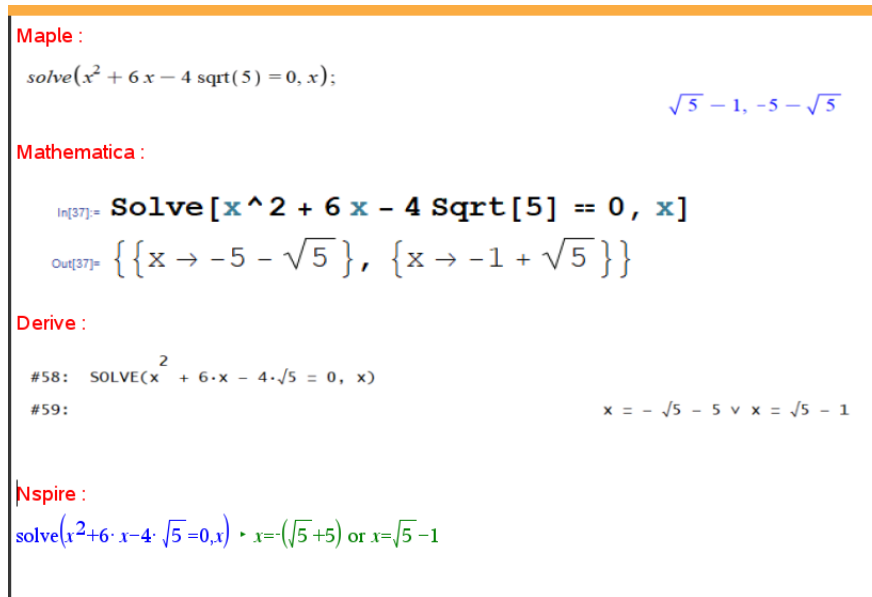


Figure 7: Using a CAS for solving a quadratic equation

and doing things like the followings : evaluate the function f at many different points (x, y) , plot a 2D graph and replacing the variable y by a slider and, finally, plot the 3D graph of it. In any case, it *seemed* to be identically zero !

4.2 We were always highly impressed by so many simplifications

Josef was not very familiar with complex analysis but knew that when a computer algebra system uses a notation like $z^{1/n}$ where n is a positive integer and z a complex number, we are (usually) talking about the **principal branch** of $z^{1/n}$: the point in the complex plane whose magnitude is $|z|^{1/n}$ and whose phase is the phase of z divided by n . For example, the angle of -1 is π .

As we noted, Josef was a slider lover! Sliders have become a fantastic tool for teaching. Of course, we can't use a slider to prove a formula. But sliders play of role of investigation and Josef used a lot of it.

4.2.1 Example 8 : everything allowed but try do show it

Here are two different problems that are related in the sense that some computer algebra system can find a quite simple answer to the question while others don't succeed.

First question : perform the following test using your favorite calculation system. Ask it to calculate the following number by answering in rectangular form : that is in the form $a + bi$ where a and $b \in \mathbb{R}$ and where i stands for the imaginay unit. The problem is : show that

$$\frac{(-1)^{1/8}}{2^{3/4}} \sqrt{1+i} + i e^{\pi i/2} = \frac{1}{2} + \frac{i}{2}.$$

Second question : for any $r \in \mathbb{R}$, show that

$$\int_0^{\pi/2} \frac{1}{1 + \tan^r(x)} dx = \frac{\pi}{4}.$$

Since r is not necessarily an integer, the calculation should most likely start by replacing $\tan^r(x)$ with $\exp(r \ln(\tan(x)))$. The nice surprise, however, is one concerning a property of the definite integral. For his part, Josef, using a graph, a cursor and graphical analysis, seemed convinced of the result.

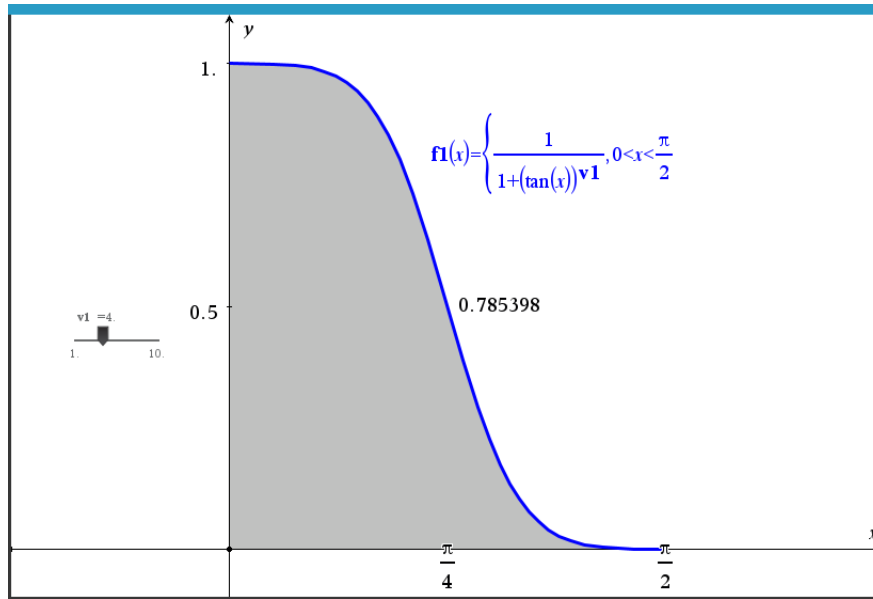


Figure 8: Using a Nspire with a slider

4.3 Together we learned using different computer algebra systems

Even though Derive had been discontinued in 2007, Josef was probably the person on Earth who continued to use it the most. He was aware, however, that with new versions of Windows, Derive's interface would become increasingly difficult to keep up. For this reason — and also because he quite enjoyed Nspire — Josef liked to cross-check his answers obtained by Derive with those given by Nspire. We conclude this article with another example Josef sent to me and showing the importance for the teacher of emphasizing the famous constant of integration, so often hidden!

4.3.1 Example 9 : everything allowed but try do show it

Here is an easy double integral from a multiple variable calculus course that students should be able to compute by hand :

$$\int_0^1 \int_1^2 \frac{xy(x^2 + y^2 - 1)}{2} dy dx = \frac{3}{4}.$$

The domain of integration is a rectangle but the integrand is not separable, so we first compute the inner integral. The funny thing is when you do it by hand, starting to compute the inner

indefinite integral. As students always do, they start finding an antiderivative for the indefinite integral. So here are one way to do this. In the inner integral, you can factor out $\frac{x}{2}$ and set $w = x^2 + y^2 - 1$, so $dw = 2y dy$ and

$$I = \int y(x^2 + y^2 - 1)dy = \int \frac{w}{2}dw = \frac{(x^2 + y^2 - 1)^2}{4}.$$

But if you avoid using a substitution and distribute the factor y over the parenthesis, you find

$$I = \int (x^2y + y^3 - y)dy = \frac{x^2y^2}{2} + \frac{y^4}{4} - \frac{y^2}{2}.$$

Don't forget, the inner integral was compute with respect to the variable y , so x was considered as a *constant*. Are both answers for I the same ? No, they differ by a ... *constant*!

$$\frac{(x^2 + y^2 - 1)^2}{4} - \left(\frac{x^2y^2}{2} + \frac{y^4}{4} - \frac{y^2}{2} \right) = \frac{x^4}{4} - \frac{x^2}{2} + \frac{1}{4}.$$

5 Conclusion

All these wonderful conferences or email exchanges during which colleagues interacted with me and showed me lots of interesting things, all these email exchanges where, receiving answers to questions I had asked or receiving answers to my questions, pretty much sum up my 30 years and more of teaching mathematics to future engineers at ÉTS. Thanks to Eugenio, Albert, Josef and so many others.

The question posed by Eugenio was a golden opportunity for me to look at how to develop our lesson plans and to try to answer the question of what our students should minimally know now that so many tools are available: my recent presentations at the ACA2021, 2022 and 2023 conferences (Using CAS in the classroom: personal thoughts, Part I, Part II and Part III) were in this direction.

For a good number of years now, I have completely disagreed with the way integration techniques are taught to students. I already made this clear during my presentation at ACA2011. This was thanks to Rubi whose development was well underway at the time. And my reflection continued during each of my presentations at ACA2024.

In all modesty, I don't think that without the almost daily use of computer algebra – and the fact that the students sitting in front of me have the opportunity to use a symbolic calculator to do and learn mathematics — that the enthusiasm would still be there after all these years. I'll preach for my parish in closing: thank you ÉTS, thank you Derive, thank you Rubi, thank you Nspire. A person like Josef Böhm has played an important role in reminding us that all these high-performance computing systems (CAS, Wolfram Alpah, AI) should not make us lose the pleasure of teaching mathematics.

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